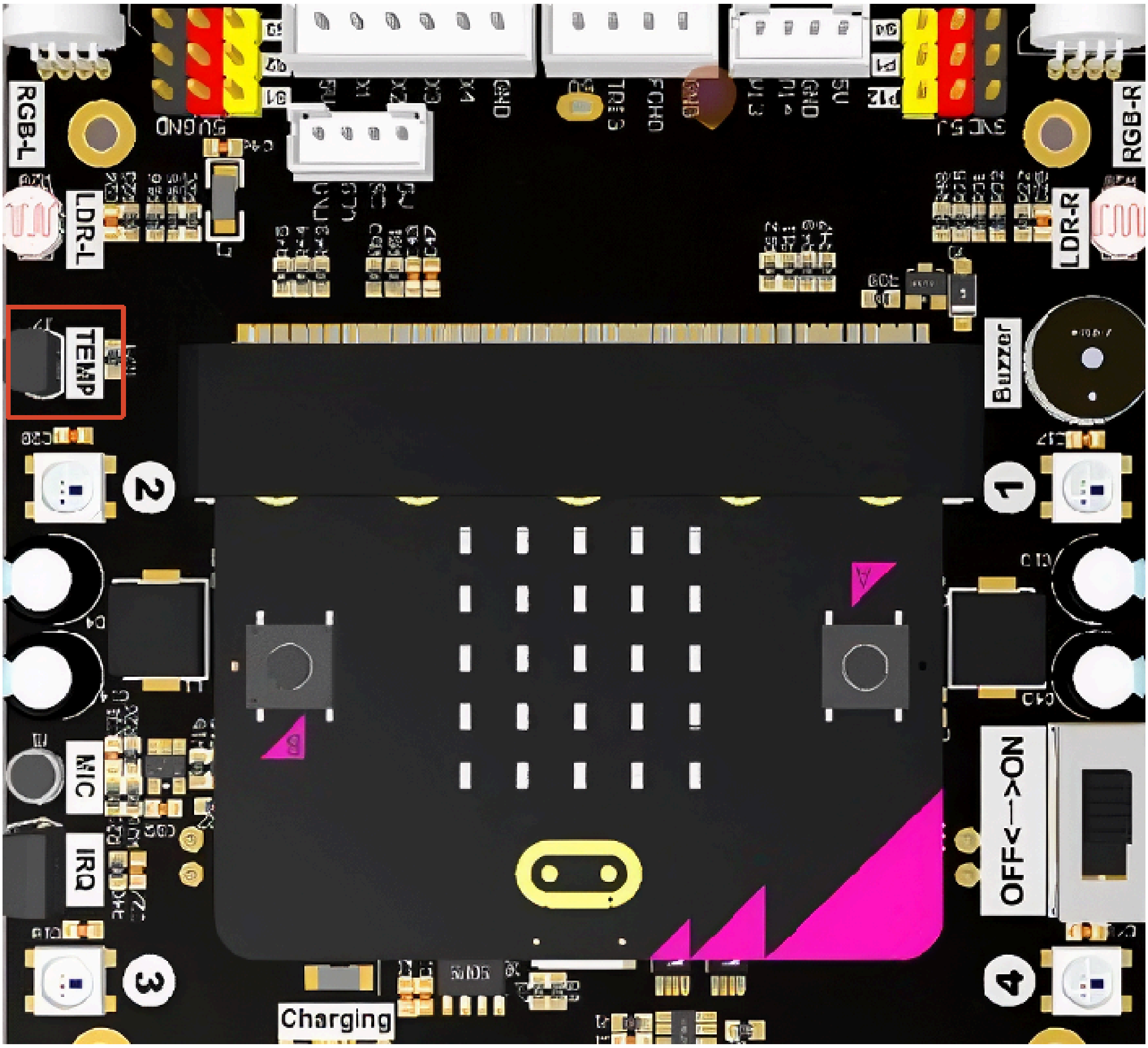
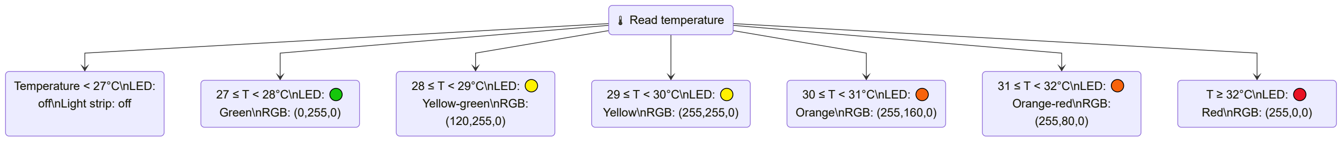
Thermometer Monitoring

Function Description

In this section, we use the temperature sensor to detect ambient temperature and display the current temperature status in real time through changes in the LED light color.

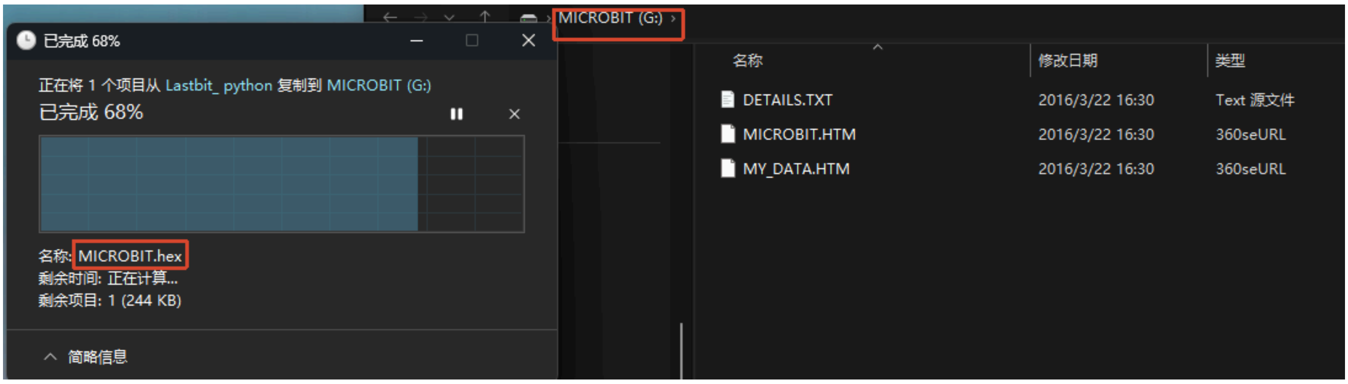


Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

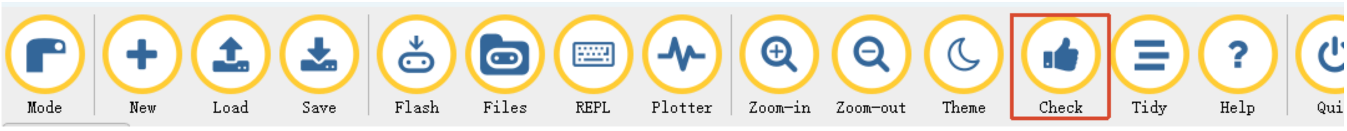
   Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will

notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

4. Now you can click `Check` to verify whether there are any problems with the code format. This case involves multi-line display patterns that may trigger warnings, but this does not affect flashing or running!



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.

6. Unplug the microUSB data cable and switch the switch to the ON position.

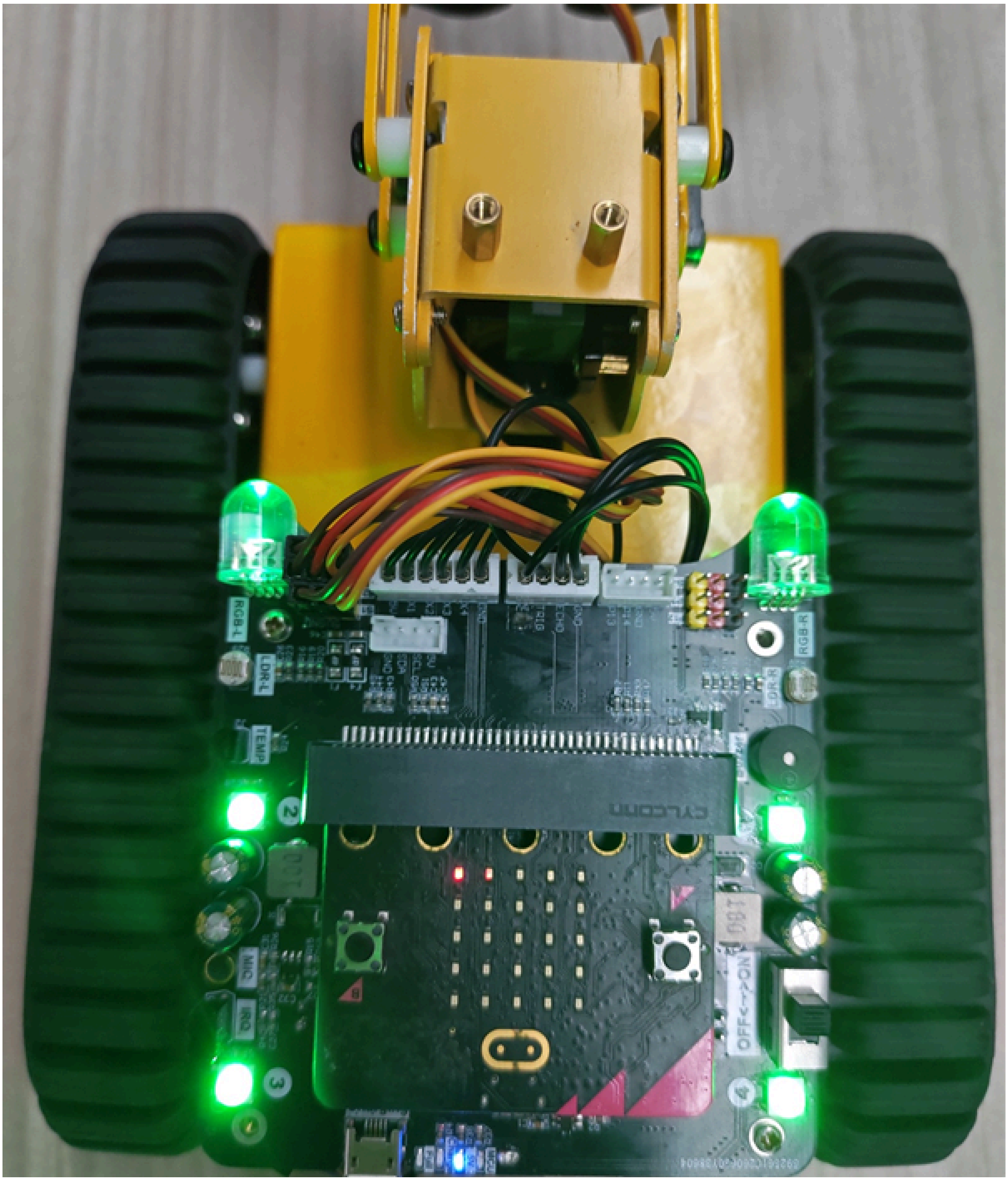
Code Implementation

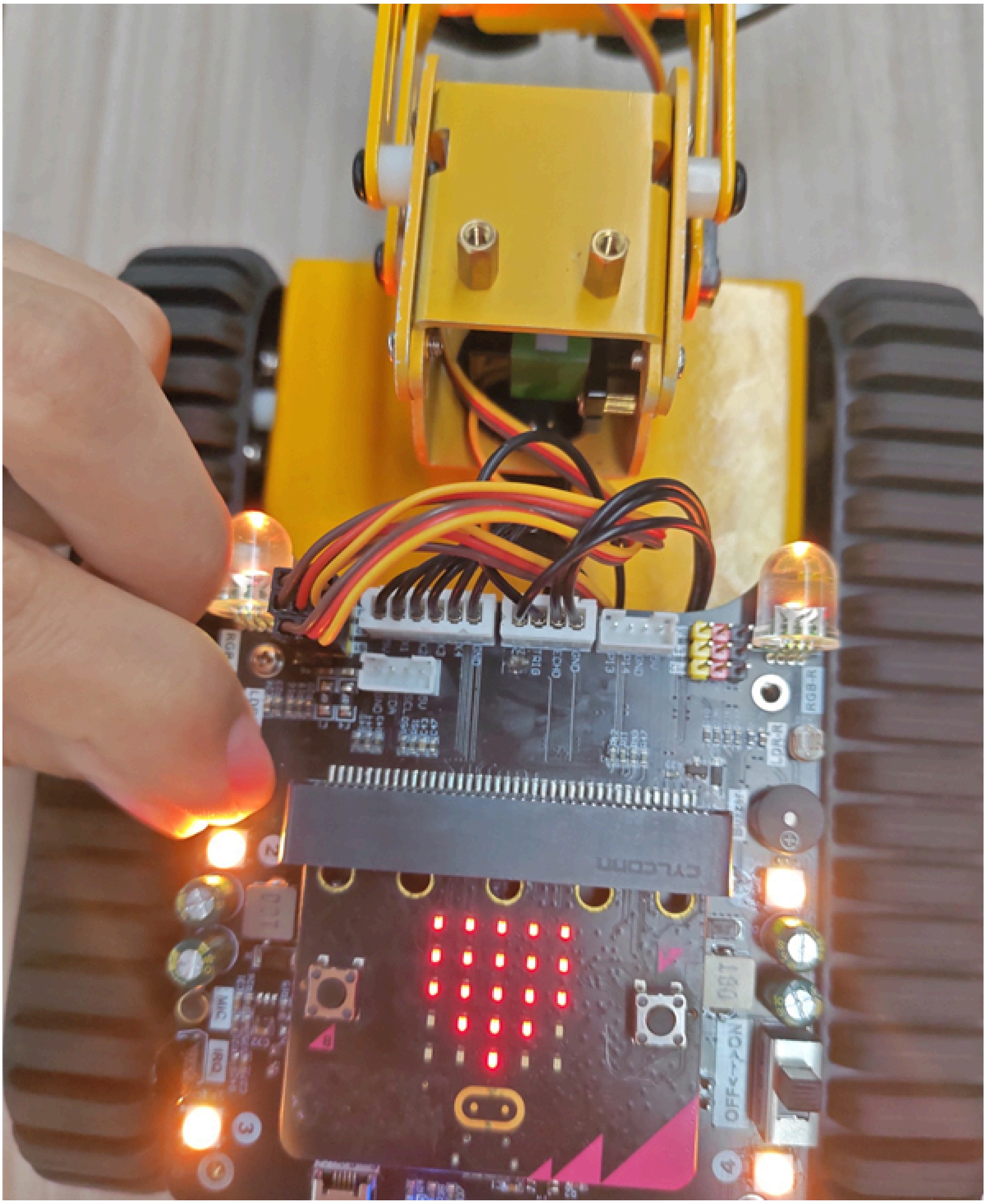
```python
# 20. Thermometer Guardian
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()
Lastbit.stop_motor()
# Current temperature
temperature = 0

while True:
    # Read temperature
    temp = Lastbit.read_temperature()
    if temp is not None:
        temperature = temp

    # Display different effects and colors based on temperature
    if temperature < 27:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "00000:"
                           "00900"))
    elif temperature < 28:
        display.show(Image("00000:"
```

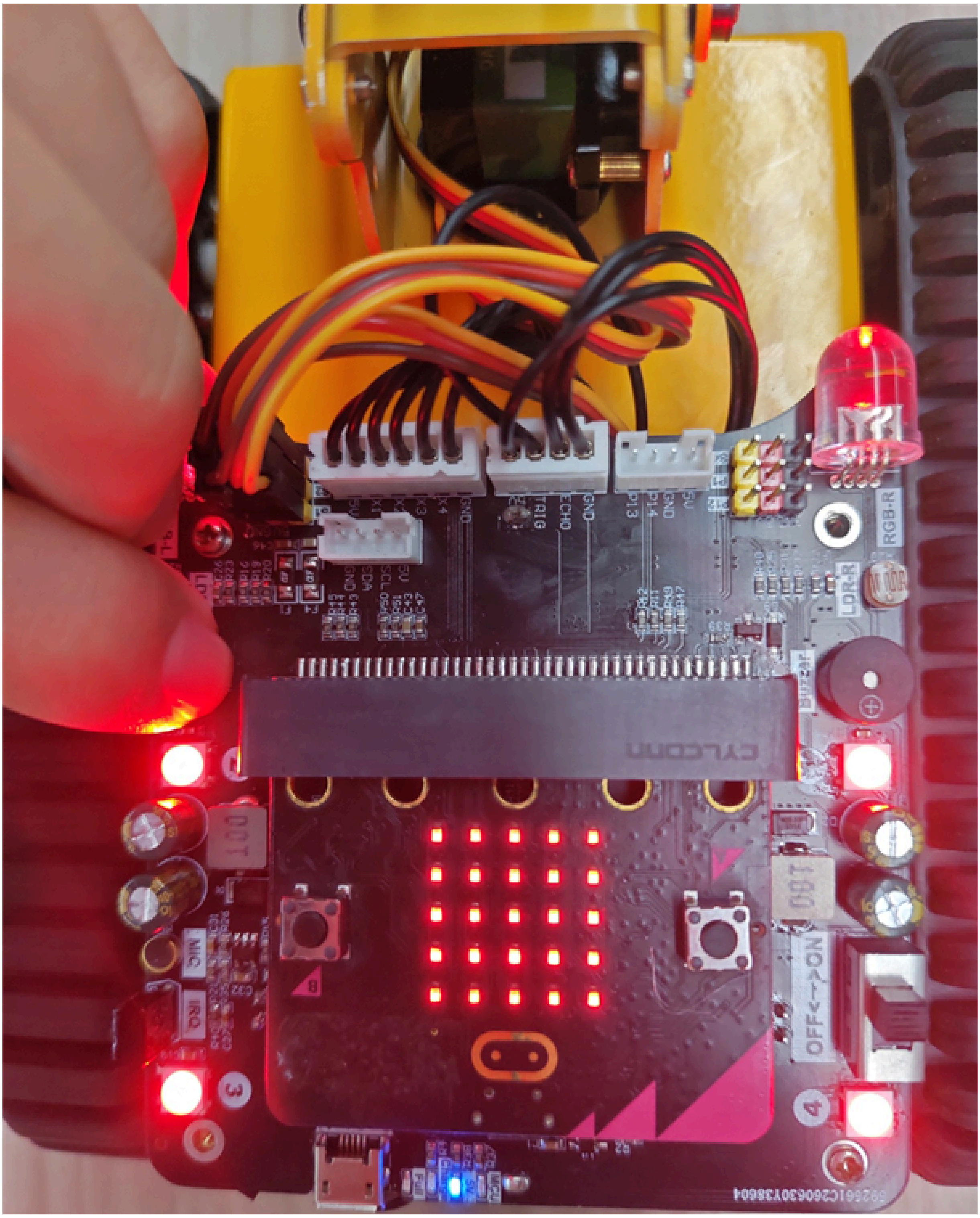
```
                         "00000:"
                         "00000:"
                         "00000:"
                         "99999"))
        # Green
        Lastbit.all_lights_set_color(0, 255, 0)
    elif temperature < 29:
        display.show(Image("00000:"
                         "00000:"
                         "00000:"
                         "00900:"
                         "99999"))
        # Yellow-green
        Lastbit.all_lights_set_color(120, 255, 0)
    elif temperature < 30:
        display.show(Image("00000:"
                         "00000:"
                         "00900:"
                         "09990:"
                         "99999"))
        # Yellow
        Lastbit.all_lights_set_color(255, 255, 0)
    elif temperature < 31:
        display.show(Image("00900:"
                         "00900:"
                         "99999:"
                         "99999:"
                         "99999"))
        # Orange
        Lastbit.all_lights_set_color(255, 160, 0)
    elif temperature < 32:
        display.show(Image("00900:"
                         "09990:"
                         "99999:"
                         "99999:"
                         "99999"))
        # Orange-red
        Lastbit.all_lights_set_color(255, 80, 0)
    else:
        display.show(Image("99999:"
                         "99999:"
                         "99999:"
                         "99999:"
                         "99999"))
        # Red
        Lastbit.all_lights_set_color(255, 0, 0)
```

Experimental Phenomenon

Make sure your fingers are dry. Contact with water may cause a short circuit on the main board! After flashing the program, you can rub the temperature sensor with your hands and notice the temperature gradually rising. The car lights will gradually change from green to orange to red. At the same time, the higher the temperature, the higher the pattern displayed on the matrix screen. After releasing your hands, the temperature sensor data will slowly decrease, and the lights will gradually change from red to orange to green.






TRIG
ECHO
GND
P13
P14
GND
5V
RGB-R
LDR-R
OFF<-->ON
MIC
IRQ